Agriculture mobility and technology networks

A connected research atlas for Blackwish

Evidence, comparison and original design proposals

21 October 2026 · Reader 1

Research and original design proposals only. Not ratified or deployed.

The largest section compares large game systems with real production and service networks. Agriculture, husbandry, genetics and transport then supply the conditions those networks must represent.

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Source register

Numbered source links lead to the relevant source register entry. Linked source titles open the original reference. Original proposals and toy examples are labeled separately from evidence.

CHAPTER 01

How to read this atlas

The useful unit is a working capability, not a civilization rank.

The central answer

Design reading

Agriculture, animal care, transport and manufacturing fit together as networks of continuing work. A crop sequence changes the timing of demand and replenishment. A cart changes which materials can move, but only along suitable routes. A breeding population carries variation, not a universal upgrade score. A design can be known while the tools, permissions or maintenance needed to use it remain unavailable.

For Blackwish, the strongest model is therefore a capability network with different kinds of links: a method requires something, an alternative substitutes for it, a process consumes or replenishes a stock, a service maintains another service, and evidence supports a claim. Historical precedence, evolutionary descent and gameplay prerequisites must not share one undifferentiated arrow. This is a proposed architecture, not an adopted Blackwish rule.

The largest part of this atlas compares technology models, including large games and concrete real-world systems. The following chapters supply the biological, seasonal and logistical detail that a technology screen usually hides. Read them together: fodder connects crop rotation to husbandry; husbandry connects to traction and transport; transport changes access to food, repair and exchange; genetic

taxonomy tells us which biological relationships are actually being represented.

Evidence and scope

Design reading

Three labels separate different jobs. Evidence reports a documented source or an explicitly bounded interpretation. Design reading explains a useful comparison without claiming it is a historical law. Original proposal introduces an untested Blackwish mechanism or fictional example. Source links sit beside each evidence passage; the source register identifies the scope of the underlying material.

This is selected comparative research, not a complete global chronology, a site-specific farming plan, veterinary advice, or a calibrated simulation. Calendar dates and game editions belong to their examples. No live system, farm treatment, breeding program or human experiment is authorized by the atlas. Proposed methods remain subject to explicit manual ratification.

CHAPTER 02

Technology trees and capability networks

A large tree becomes more useful when it shows alternatives, production, access and maintenance rather than merely adding more unlocks.

Four graphs that should not be confused

Design reading

A research graph represents what must be learned or researched under a particular ruleset. A production graph represents how materials and energy become outputs. A service graph represents the people, infrastructure and access that keep a capability usable. A historical graph represents documented sequences, exchanges and changes, with uncertainty. Games often combine these for clarity, but their arrows answer different questions.

Consider a fictional water mill. Knowledge of gearing is not the same as a built wheel; a built wheel is not the same as a dependable stream; a usable stream is not the same as permission to divert it. Flour also requires grain, a sound millstone, maintenance and distribution. A single mill icon can be a helpful interface, but a deeper model should preserve the underlying requirements and alternatives.

Research
What is learned or demonstrated?

Production
What becomes what, using which inputs?

Service
What keeps the capability available?

Evidence
What supports the claim, and where?

Original schema. Research, production, service and evidence are separate layers joined by explicit typed links.

Real systems do not arrive as complete unlocks

The Computer History Museum recounts building Difference Engine No. 2 from Babbage’s 1847-49 designs. The original drawings did not provide every manufacturing detail, so additional detailed drawings were needed. Modern manufacture was used while respecting precision achievable in Babbage’s era. The lesson is narrow but useful: possessing an ingenious design does not prove that a complete production, financing and operating system was historically available. The Difference Engine and Analytical Engine are distinct designs. [1]

NASA’s Technology Readiness Levels measure the maturity of a particular technology. They are not a ranking of societies, usefulness or ecological desirability. For this atlas, readiness is a prompt to separate demonstration from routine operation, not a scale to assign to a whole culture. [2]

Design reading

A practical model can ask six separate questions: Is the capability understood? Has it been demonstrated? Can this actor produce or obtain it? Is it working now? Can it be maintained? Is it suitable for the present goal? These questions allow a technically sophisticated society to retain simple tools for appropriate tasks without being portrayed as having failed to progress.

A real catalogue with 1337 entries

Wu and colleagues catalogued 1,337 ancient Chinese agricultural technologies across a study period from 8000 BCE to 1911 CE. Their four-level classification includes theories, engineering, practices, protection and crops. Its scope focuses on crop-related land and water mobilization, excluding husbandry and several other agricultural domains. This is a large historical catalogue, not an empirically proven prerequisite graph. Entries were coded from historical encyclopedias, and classification and preservation influence what is counted. The authors identify network analysis as further work; diffusion direction can remain unresolved. Its value here is scale and organization, not a universal growth curve or a ranking of cultures. [3]

The OECD and Eurostat Oslo Manual distinguishes implementation from invention and discusses capabilities and knowledge flows. Skills and organizational capacity are not equivalent to possession of information, and diffusion does not ensure adoption. These distinctions are useful here, but modern business-survey indicators should not be used as direct measurements of ancient communities. [4]

A large agricultural capability network

Design reading

The following is an original map of 48 capability nodes. It is deliberately not a historical sequence. Each row is a cluster of interacting capabilities; entries are not claims that every farm needs every item. The point is to expose where a large tree can earn its size by describing meaningful differences rather than splitting one generic farming bonus into dozens of near-identical upgrades.

Original proposal

Observation and records. Seasonal observation; field histories; soil and water assessment; crop identification; animal identity; storage inspection.

Plant production. Seed or planting stock; propagation; seedbed preparation; sowing or transplanting; weed management; harvest coordination.

Soil and water. Erosion protection; organic cover; nutrient budgeting; nitrogen-fixing associations; irrigation delivery; drainage and salinity assessment.

Animal care. Species-appropriate nutrition; reliable drinking water; shelter and social conditions; skilled handling; health surveillance; reproductive care.

Postharvest systems. Drying; preservation; containers; protected storage; milling or processing; safe byproduct handling.

Materials and power. Tool materials; joining and fabrication; traction; rotary power; repair skills; replacement parts.

Movement and exchange. Human carrying; pack systems; carts and routes; boats and landings; transfer and loading; access and exchange arrangements.

Resilience and governance. Shared calendars; water allocation; resource access; reserves; contingency care; evidence review and revision.

Design reading

The network is large because it connects distinct functions. A storage node can matter to a mobile household, a monastery farm, a rice-fish system or a factory town. A seasonal calendar can constrain all of them without being a biological trait. A field may have a viable water supply yet lack the labor available at harvest. The display should let the reader follow those cross-links instead of forcing each capability into one era.

Three worked production chains

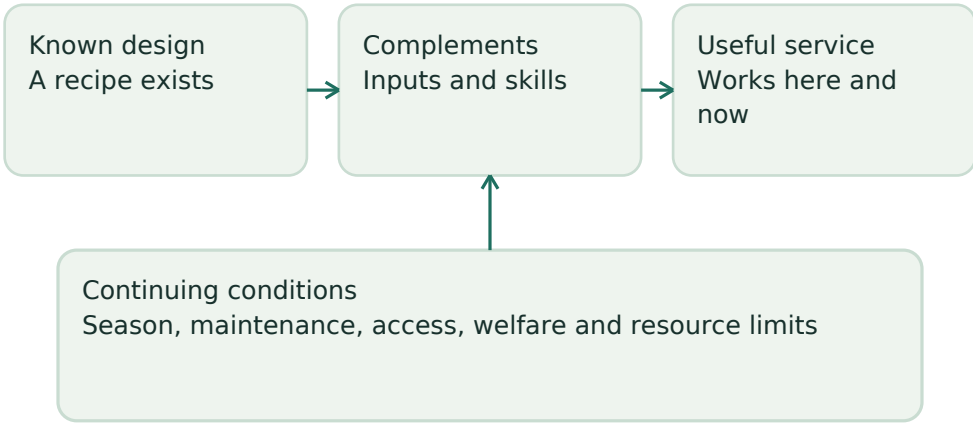
Design reading

Food chain, original illustrative model: planting stock plus a viable field and seasonal work produce a crop; harvest and preservation convert it into usable food; containers, transport and access determine who receives it. A portion may be reserved for seed or feed. Residues may protect soil, feed animals, become bedding or supply fuel, but cannot be allocated in full to all uses. A healthy harvest does not prove that the food reached households.

Metalwork chain, original illustrative model: access to appropriate raw material, energy, equipment and skills enables a particular material-processing recipe. The output may be semifinished stock rather than a finished implement. Fabrication, joining, inspection and maintenance remain separate services. This schematic deliberately gives no

smelting temperatures, alloy recipes or operational instructions; it illustrates why an imported ingot is not identical to local mastery of every upstream process.

Electrified service chain, original illustrative model: a generator supplies electricity only when input energy and operating conditions permit; conversion, distribution and sometimes storage make it available to a load; controls, spare parts and repair keep the service functioning. Renewable electricity does not replenish every material it powers. A solar water pump can therefore improve energy supply without establishing a sustainable water budget.



Original toy chain. One useful service can fail through material shortage, seasonal unavailability, maintenance loss or lack of access while the design remains known.

Thirteen large game cases and comparators

Design reading

The selection below includes twelve substantial progression systems plus Freeciv as an inspectable implementation comparator. Large means that several important systems interact, not an unsupported claim about the exact number of nodes. The examples deliberately span civilization strategy, science-fiction webs, factories, material crafting and institutional economics. Each case fixes its evidence to a version, manual or dated design account rather than implying that all current patches behave identically.

Civilization IV and explicit prerequisites

Scope: Firaxis/2K base-game manual, 2005. Expansion rules and later balance changes are outside this case. [5]

The manual presents both alternative prerequisites and combinations: Animal Husbandry can follow Hunting or Agriculture; Pottery combines The Wheel with an alternative route through Agriculture or Fishing. Research unlocks things that still have to be produced or used. The resource section distinguishes an improved local resource from access through a connecting transport network. Technologies can also be acquired through diplomatic exchange. These are game rules, not historical demonstrations that the named ideas necessarily depended on each other. [5]

Original design proposal: mark AND and OR requirements directly. A target might require every input in one group but any one route in another. Then separate eligibility to attempt a task from reliable delivery. Two settlements can hold identical instructions while only one has dependable supplies. Imported knowledge and an imported finished object should

also create different records: receiving a blade does not establish local manufacturing or repair capacity.

Civilization VI and learning through activity

Scope: Base-game design and official 2017 introductory guidance; Gathering Storm environmental design discussed separately; Kublai Khan DLC example dated 2021. No fixed Eureka percentage is asserted because rulesets and abilities differ. [6] [7] [8]

Official introductory guidance distinguishes Science and Culture unlocking technologies and civics. Particular actions accelerate research: settling a coastal city boosts Sailing, while defeating a unit with a Slinger boosts Archery. These examples show an action-to-research link, not the complete cost or prerequisite rules. [6]

The official overview describes physically expanding cities and active research in both Culture and Technology. Its Gathering Storm section adds floods, droughts, storms, and development choices that affect global temperature, ice, and sea level; that expansion also extends both progression trees. [7]

The 2021 publisher announcement gives Kublai Khan a random Eureka and Inspiration when first establishing a trading post in another civilization's city. This is a particular leader ability, not a universal rule for all trade. [8]

Original design proposal: record practical trials, observations, inherited craft and organizational learning as different evidence channels. One successful action can raise confidence without certifying year-round operation. A contact event can create a learning opportunity while translation, training or adaptation remains unfinished. Environmental feedback should change feasible choices rather than appear only as a penalty detached from the production chain.

Civilization VII and historical chapters

Scope: Original prelaunch Ages explanation by the creative director. This is a historical design snapshot, not a statement of every rule in the current 2026 release. [9]

The original design organizes play into Antiquity, Exploration, and Modern Ages, each controlling available technologies, civics, resources, units, and civilization choices. Transitions preserve selected legacies and enduring elements while changing the playable context. The developers present the system partly as a response to late-game snowballing and engagement problems. [9]

Original design proposal: use scenario chapters to change the level of detail, not to erase history. A village view and an interregional view can show different parts of the same underlying network. Preserve continuity in tools, specialists, records and routes. A political breakup can coexist with continuing craft and trade; an interface transition, a political transition and a service interruption are not the same event.

Beyond Earth and lateral research

Scope: 2014 base-game extended manual. Rising Tide additions and later balance are not silently included. [10]

The manual describes a radial web whose connections can be traversed in either direction from known technologies. Broad branch technologies provide the backbone; attached leaf technologies focus on applications and contribute to Affinity development. Thematic threads permit lateral movement without backfilling unrelated research. Science costs and connection rules still constrain progression. The manual's ratios are not used here because its numerical descriptions are not internally consistent enough to support a reliable total. [10]

Original design proposal: distinguish a general capability family from specific applications. Permit lateral transfer when a technique genuinely informs another use, with the transfer mechanism named. A community may gain a narrow application through contact without mastering every neighboring theory. Specialization can therefore be viable while leaving gaps in repair or alternative supply. The web should reveal those consequences rather than merely reward completing everything.

Alpha Centauri and industrial capacity

Scope: 1999 base-game manual. The source is an original manual preserved by an independent archive, not a newly verified publisher-hosted download. Alien Crossfire is outside this case. [11]

The manual explains that technology represents both scientific understanding and the organization needed for a self-sustaining industry. Its ocean-going-fleet example does not mean the settlers have forgotten how boats work. Research uses labs and prerequisites. Under default blind research, the player sets priority categories rather than selecting an exact next discovery; a setup option permits directed research instead. [11]

Original design proposal: preserve inherited knowledge during migration or disaster while separately rebuilding equipment and coordination. A shipwrecked group can retain navigation books and experienced sailors but lack a shipyard or suitable vessel. An individual expert may demonstrate a method without establishing a durable training system. Record what the node means so a familiar label does not silently alternate between an idea, a skill and an entire industry.

Factorio and research as production

Scope: The 2023 developer diary describes changes planned for Factorio 2.0 and Space Age. Official wiki entries were checked on 2026-10-21. Base-game and expansion space-science mechanics are explicitly separated. [12] [13] [14] [15]

The developers identify a bootstrap problem: a planet's science pack should require its processing chain, but that chain must be learned before the pack exists. Trigger technologies therefore respond to actions such as mining, crafting, or launching particular material. The diary uses contact with crude oil and actually producing steel as examples. Science-pack research remains important at scale. It also explains quality as optional production complexity whose random results can average into production ratios at large output. [12]

Science packs are physical products consumed by labs. Research therefore depends on supplying the required pack types, not merely owning a laboratory. Different pack-production times imply different machine allocations for balanced throughput. The documentation distinguishes the base game from expansion additions. [13]

Base-game space science is obtained through satellite launches. In Space Age, it becomes an earlier platform-based production process using collected asteroid materials. This is a concrete warning that an identical technology-item name can occupy a different place in different editions. [14]

Space Age spoilage continues while relevant items move through belts, machines, and storage. On Gleba, nutrients and other perishable materials connect production to time. A fallback conversion from spoilage to nutrients helps bootstrap production. Perishable research inputs make transport delay matter as well as quantity. [15]

Original design proposal: distinguish producing one sample from sustaining throughput across a season. Check bootstrap loops explicitly: if a process needs an input whose only source requires that process, provide a starter stock or independent route. Track rates and travel time as well as inventories. A full store can conceal depletion, and additional laboratories cannot solve an upstream shortage. Keep inefficient but recoverable fallback methods visible after disruption.

Satisfactory and alternative recipes

Scope: Coffee Stain's 1.0 release material from September 2024 and official wiki documentation as accessed in 2026. Exact current recipe totals, hard-drive pools, and balance coefficients are not asserted. [16] [17]

The 1.0 announcement includes Project Assembly's final phase and Tier 9, with advanced production connecting earlier components to new machinery. Its hard-drive system keeps researched options in a library and allows a reroll of the offered choices, rather than forcing immediate selection before other research. Late production includes processes with linked input and byproduct streams, such as the Quantum Encoder. These are fictional game processes. [16]

The official community wiki describes alternative recipes as exchanges among material use, energy, machinery, space, complexity, and available resources. They are not uniformly superior replacements. Their value depends on the surrounding factory and local supply situation. [17]

Original design proposal: compare alternatives as a vector of labor, land, water, fuel, materials, repairability, quality, waste and reliability. A labor-saving mill dependent on imported parts need not dominate a locally repairable method during interruption. Project delivery demonstrates integration once, not indefinite sustainable supply. Include coproduct handling

and waste treatment in the network so a useful output cannot conceal an unresolved disposal bottleneck.

Dyson Sphere Program and spatial logistics

Scope: Developer/publisher product description plus dated 2021-2024 announcements. The game's science-fiction premise is not a feasibility assessment of real Dyson structures or interstellar logistics. [18] [19] [20] [21] [22]

The official product description presents an interstellar production network across planets with different resources. Factory growth and research are linked, and blueprints support reuse of production layouts. A 2021 patch explanation notes that more Matrix Labs can accelerate research; this does not remove input requirements. [18] [19]

The 2024 logistics development diary describes supply-demand pairing between stations and explains why a mechanically valid route can still produce undesirable long-distance shipments. New priority dimensions distinguish explicit station pairings, planetary or stellar routes, groups, and general matching. Feasible service also depends on cargo, energy, and transport equipment. [20]

The November 2024 statistics update adds upstream/downstream recipe inspection and ideal production/consumption references. This makes intended production structure and theoretical rates easier to inspect; actual delivery still requires a working network. [21]

The 2022 metadata feature derives a cross-save resource from matrix production and allows technology buyout in other eligible cluster-address saves. It is meta-progression across playthroughs, not ordinary within-world diffusion between societies. [22]

Original design proposal: locate every stock and service. Distinguish planned output, nominal capacity, observed production and useful output delivered to the right destination. Aggregate supply can be adequate while allocation and delays leave a critical workshop unsupported. Blueprint transfer moves information, not equipment. Make route priority and access legible so the network can explain who receives scarce resources rather than reporting only a reassuring global inventory.

Stellaris and uncertain opportunities

Scope: PC developer material for 3.7 and 3.9, plus a separately labeled 2025 discussion of proposed 4.0 changes. Console editions and later patches may differ. The original 2015 research diary could not be fully retrieved and is not used as the evidentiary basis for precise current rules. [23] [24] [25]

The 2023 archaeotechnology diary describes research options gained through precursor projects and excavations. The resulting buildings, modules, and ship components require minor artifacts to build. Excavated sites can provide artifact deposits. The Fear of the Dark origin reduces available research alternatives. These mechanics distinguish opportunity, knowledge, and material deployment. [23]

The official 3.9 announcement says Council Agendas can grant partial research toward technologies needed for chosen ascensions, reducing dependence on random availability. This is a documented way of steering around uncertainty, with DLC conditions. [24]

A 2025 4.0 development discussion explicitly refers to random pulls from a technology deck and proposes milestone-based guaranteed options for important technologies. It documents a design direction, not proof that every proposed detail shipped unchanged. [25]

Original design proposal: separate possible, locally known, currently investigable and selected. Finding an artifact may prove that a mechanism existed without revealing its alloy or tolerances. Allow deliberate investigation and training to reduce uncertainty instead of making an essential service depend solely on a random menu. Keep rare material access independent of understanding, and never borrow fictional biological ascension as a model of human taxonomy.

Age of Empires II and visible specialization

Scope: Official Age of Empires II: Definitive Edition teaching material, with the 2025 Chronicles: Alexander the Great explanation treated as a distinct mode/content family. [26] [27]

The official tutorial explains how the technology-tree display differentiates buildings, units, and technologies, and marks unavailable options for a selected civilization. Access to unit lines and upgrades varies between civilizations, making shared architecture and asymmetric availability visible. [26]

The 2025 Chronicles explanation describes familiar unit and technology roles with ancient-themed names and art. It also distinguishes Chronicles' paired unique-technology choices: players select one option in each of the final two Ages. This customization should not be generalized to every ordinary AoE II civilization. [27]

Original design proposal: use distinct visual forms for a building, practice, product and policy. Explain exclusions through circumstances or explicit scenario rules, not permanent ancestry-based ceilings. Unavailable now should differ from impossible forever. A migrating group may acquire neighboring techniques without changing its biological ancestry; a short-horizon strategic commitment should not masquerade as a law governing centuries of cultural change.

Vintage Story and embodied craft

Scope: Officially hosted, community-maintained wiki checked on 2026-10-21. Forge documentation includes 1.22-era changes. Exact timings, temperatures, recipe quantities, and wear probabilities are intentionally omitted. [28] [29]

Steelmaking carburizes iron in a cementation furnace, produces blister steel, and then refines it. The documented game process needs an appropriate anvil, forge, bellows, pulverizer, and a multiblock furnace. Refractory bricks depend on selected minerals; higher grades reduce replacement needs. Firing can damage refractory blocks, so repeated production requires repair materials. [28]

Forge use depends on fuel and sufficient working heat. The current documentation's bellows requirements illustrate that material-processing details can change between versions. The atlas therefore uses the process relationships, not an asserted timeless game recipe. [29]

Original design proposal: divide a material capability into feedstock, energy, containment, control, forming, finishing and assessment. Separate construction inputs from maintenance inputs. A workshop can retain skilled people and records while production stops because replacement lining is unavailable. Reliability and local repair can be more important than peak output. Borrow the structure of dependencies, not the game’s literal recipe as historical metallurgy.

Victoria 3 and institutional adoption

Scope: Prelaunch developer diaries from 2021. These establish design concepts and examples, not every current employment, ownership, literacy, or diffusion formula. [30] [31]

The Technology diary divides research among Production, Military, and Society. Funded universities employ people and

increase Innovation; literacy limits directed use of that research capacity. Knowledge also spreads from elsewhere, affected by literacy and speech rules. Technologies usually open options rather than automatically transforming production. The diary distinguishes research from the supporting inputs needed to use an invention and notes that education affects future workforce qualifications. [30]

The Production Methods diary defines methods through their inputs, outputs, and worker types. New methods need not be superior in every circumstance: they can require unavailable goods or exchange one output for another. The original developer text is available through preserved public reproductions; the direct Steam page currently exposes little text, so this source is used only for those broad design statements. [31]

Original design proposal: represent knowledge holders, training and opportunities to practice. A past generation’s achievement should not guarantee an eternally staffed service. Diffusion needs its own links and delays, while adoption remains a choice with material and distributional effects. A machine can increase total production yet threaten access or depend on insecure supplies. Such resistance need not be irrational, and more aggregate output does not establish a fair transition.

Freeciv and inspectable implementation

Scope: Official project website checked on 2026-10-21. No particular ruleset, technology total, or exact source-code behavior is claimed here. [32]

The official project identifies Freeciv as free and open-source empire-building strategy software and links its public repository. It states that Freeciv is released under the GNU General Public License, while the Freeciv-web client uses the

GNU Affero General Public License. Those are distinct licensing statements, not permission to treat every related mod or asset as public domain. [32]

Original design proposal: if implementation analysis is later needed, pin a version and ruleset before counting nodes. Inspectable records make reachability, dead ends and contradictory prerequisites testable. Open implementation improves auditability, not historical truth: a transparent fictional dependency is still fictional. No code, ruleset data or artwork is imported here, and any later reuse needs file-specific provenance and licensing review.

Fictional worlds as deliberate rule choices

Failbetter presents Sunless Skies as a Gothic horror RPG with a Victorian empire, a spacefaring steam locomotive, exploration and locomotive upgrades. Its cosmology contains explicitly impossible premises. It is a useful steam-era fictional example, not evidence that historical steam engineering supports space travel. [33]

Solarpunk Magazine describes an aspirational movement imagining sustainable, more balanced societies through speculative art. Its framing allows conflict and unfinished problems. This is one participant publication’s account, not a binding genre definition. Optimistic green imagery alone does not demonstrate renewable-resource sufficiency. [34]

ION LANDS identifies Cloudpunk as a story-based cyberpunk game about delivery driver Rania in the vertical city of Nivalis, explored by hover car and on foot. The verified reference is the named game, not a standardized independent cloudpunk genre or a historical stage. Its mobility is fictional. [35]

Design reading

Original design reading: a steampunk-inspired network can make precision workshops, pressure-vessel maintenance, fuel supply and social access important while clearly labeling any impossible physics. A solarpunk-inspired network can emphasize repair, ecological limits, distributed services and fair access without turning renewable energy into unlimited materials. A Cloudpunk-inspired logistics setting can distinguish vertical distance from travel time, ownership from route permission, and a vehicle from a reliable delivery service.

What to borrow from large games

Design reading

Borrow a form because it serves a question. A visible tree helps a reader plan. A web supports strategic specialization. A card draw can make discovery uncertain. An action trigger can make practical achievement matter. A production dependency makes an unlock physically meaningful. An institution or workforce requirement makes adoption more than a research payment. None is automatically more realistic in every domain.

For a Blackwish interface, use a compact overview with expandable recipe and context panels. Let a player see the immediate bottleneck first, then inspect material flows, seasonal limits, alternatives and supporting evidence. Large does not have to mean unreadable. A meaningful 200-node network should distinguish 200 useful states or choices, not merely require 200 clicks.

A comparison should also separate exploration from completionism. Does the player need every branch? Can knowledge spread without being locally reinvented? Can an

obsolete process remain economical or culturally preferred? Can a known capability become unavailable? Do technologies create new obligations or unequal access? These are design questions to decide explicitly, not facts that can be copied from an archaeological timeline.

CHAPTER 03

Crop rotation through time

A rotation changes what happens on the same land over time. Its effects come from biological mechanisms, timing and the surrounding management system.

The vocabulary matters

Crop rotation is a planned sequence of different crop phases on a field. A farm can operate several fields at different phases simultaneously, so the whole rotation is represented every year. This differs from merely growing a diverse crop list somewhere on the farm. The USDA standard treats rotation as a conservation practice whose design must match local objectives and conditions. [36]

Intercropping places crops together in space and time; multiple cropping grows more than one harvest in a year. Cover crops primarily provide services such as cover or nutrient retention, while green manure returns grown biomass to the soil. A ley is a temporary grass or grass-legume phase. These practices can be combined with rotation rather than replacing it. [37] [38]

Design reading

A useful field history therefore records the actual sequence and dates, what biomass was removed or returned, inputs, weather and management. A stylized diagram is a hypothesis about functions, not proof that the functions occurred. The same cereal-to-legume arrow can describe very different nutrient exports, water use and labor demands.

Earlier agriculture was already varied

At 13 Neolithic European sites dated approximately 5900-2400 calibrated BCE, cereal and pulse isotope evidence supports early manuring and managed growing conditions. This places crop-livestock integration well before medieval field systems. It does not reveal a universal annual rotation or prove equal management intensity at every site. [39]

The La Draga study, covering approximately 5300-4800 calibrated BCE in northeastern Iberia with regional comparisons, inferred favorable water availability and relatively moderate manure inputs. It complicates a universal intensive-garden picture: early farming practices varied with setting. Isotopes and grain form are evidence to interpret, not surviving farm diaries. [40]

The Sumerian Farmer’s Instructions connects irrigation upkeep, oxen, repaired tools, sowing, harvest work, threshing and transport. Its value here is the coordinated agricultural workflow. As a prescriptive ancient text, it is not a measured trial and does not establish one exact rotation for Mesopotamia. [41]

Columella’s first-century CE Roman agronomy discusses fallow, manure, fodder, lupine green manuring and contrasting crop effects. It even records disagreement about harvested beans and fertility. Empirical management preceded modern nutrient chemistry, but an elite prescription neither proves universal practice nor supplies a safe contemporary recipe. [42]

In sixth-century northern China, Qimin Yaoshu describes mung beans incorporated before a subsequent crop and attends to timing and soil conditions. This documents a green-manure tradition outside European field systems.

Historical crop terms and yield claims require interpretation; they should not be converted directly into modern botanical identities or measured effect sizes. [43] [44]

Medieval rotation and its regional limits

In a stylized two-course system, a cropped phase alternates with fallow. A three-course sequence combines an autumn-sown crop, a spring-sown crop and fallow; the spring crop need not be a nitrogen-fixing legume. Two-course systems persisted. England’s earliest written evidence for systematic rotation is twelfth- to thirteenth-century, while earlier patterns are inferred archaeologically. Open fields concern holdings and coordination, not one mandatory crop sequence. [45]

At Stafford, evidence from plants, isotopes and animals across roughly 800-1200 CE suggests expansion, a later move toward lower-input cereal husbandry, and rotation by at least the mid-tenth century. This is a local reconstruction, not an invention date for England or a global technology unlock. [46]

The Feeding Anglo-Saxon England dataset combines plant remains, animal bones, isotopes, weed ecology and radiocarbon evidence. Its breadth matters because no single proxy fully identifies a rotation. Regional coverage and preservation remain uneven, so apparent absence can reflect what survived or was sampled. [47]

The Andalusi Books of Filaha preserve knowledge of soils, irrigation, cereals, pulses, horticulture and trees, drawing on multiple earlier traditions. Medieval agriculture thus includes intensive irrigated and horticultural branches as well as grain-fallow systems. The timing and extent of the proposed agricultural revolution remain debated. [48]

Design reading: autumn and spring are ecological windows, not universal months. A sequence useful in a cool, moist climate may fail where summer drought dominates. Shared grazing, access to residue, land tenure and the labor calendar can be as important as the species list. These relationships should be represented as conditional constraints rather than a single three-field bonus.

Other ways to organize diversity

Contemporary milpas can combine maize, beans, squash, herbs and other plants. They demonstrate spatial and food diversity, not one immutable three-species formula. A present-day description does not establish the first historical association or a uniform fallow interval. [49]

Chinampas combine constructed wetland beds, canals, tree borders and diverse cropping. They illustrate water management, organic-material handling and access working together. They are not unlimited floating soil: water quality, canal maintenance and urban pressures remain material constraints. [50]

A Norfolk-type four-course sequence documented in the Saxmundham experiment is roots, spring barley, a legume phase and winter wheat. The fully phased trial began in 1899, building on an older husbandry pattern. Fodder and fertility phases can replace uncropped fallow where conditions support their use. This is not a claim of a single inventor or universal adoption date. [51]

Modern success cases and counterexamples

At Marsden in Iowa, the 2003-2011 comparison tested maize-soybean against three- and four-year systems adding small grain, red clover or alfalfa, alongside manure and altered weed management. Maize and soybean yields averaged 4% and 9% higher in diversified systems. Annualized whole-system manufactured-N inputs averaged 80, 16 and 11 kg N/ha, respectively; these are not total N supply or maize-phase rates. Herbicide use was 88% lower. More labor was needed, and reported returns excluded land cost, management time and subsidies. The result belongs to a management package, not rotation alone. Crop-phase yield gains also do not establish an equal increase in rotation-wide edible output. [52]

The Elora, Ontario experiment followed corn-based rotations and tillage treatments from 1982 to 2012. Diversification improved corn and soybean yield stability, with responses differing by crop and weather. Small grains created windows for underseeded legumes. The result supports a context-specific resilience benefit, not a guarantee for every climate, market or measure of risk. [53]

A Central Great Plains experiment at two sites found subsequent wheat yields averaging 10% lower after cover crops than after residue-bearing fallow, with greater reductions under drier conditions. A ten-species mixture did not remove the penalty. Living cover transpires water: this counterexample defeats a universal cover-crop benefit, not all cover cropping. [54]

Broadbalk began in 1843 and provides long-term wheat, manure, mineral-nutrient and soil records. Treatments and management changed over time, so valid comparisons need

matched periods and designs. The resource is valuable precisely because a single season cannot show all long-term consequences; no numerical yield conclusion is inferred here from the experiment’s age alone. [55]

In surveyed South China rice-fish systems, rice yield and stability were similar to monoculture while pesticide and chemical-fertilizer use were 68% and 24% lower. The study also examined ecological interactions experimentally. These survey differences are not universal causal effect sizes. Fish require suitable water and care, and coculture is neither crop rotation nor a closed nutrient loop. [56]

The eight-year Salinas organic-vegetable study distinguishes total soil carbon from readily decomposable carbon, which respond differently to compost and covers. A 2024 correction replaces carbon-stock figures and tables, so uncorrected numerical estimates should not be reused. Imported compost also requires an upstream accounting boundary before making climate claims. [57] [58]

Why the sequence changes outcomes

Legume-associated fixation adds atmospheric nitrogen to biomass, but harvest may export much of it. Decomposition and residue quality govern when nutrients become available. Rotation does not manufacture phosphorus or potassium. Recycling manure and crop residues can conserve nutrients, yet exports and losses still require element-specific accounting. [59]

A nitrogen fertilizer credit is not always a net nitrogen addition. Minnesota Extension’s soybean example distinguishes the following crop’s reduced fertilizer requirement from a simplistic claim that soybeans left an

equal amount of newly fixed nitrogen in the soil. Use locally calibrated credits, not a universal legume token. [60]

Pest management works on actual hosts and life histories, not visual crop categories. Integrated Pest Management combines biological, cultural, physical and other appropriate tactics; rotation is one component rather than a substitute for diagnosis. [61]

USDA research describes corn-rootworm adaptations that weakened a long-standing corn-soybean strategy, including behavior and diapause responses. The lesson is evolutionary and specific: a repeated rotation can itself become a predictable selection environment. It does not establish current local pest pressure without current observation. [62]

Design reading

Original modeling proposal: give each phase several effects rather than one fertility number: water demand, residue amount and quality, nutrient uptake and export, host status, soil cover, work timing and market or household use. Evaluate the resulting sequence over its full cycle. Adaptive changes can be sensible, but field history must survive the change so the model does not erase accumulated risk.

CHAPTER 04

Renewable agriculture without imaginary closed loops

Prioritize replenishable flows while protecting slow stocks and accounting for losses, imported inputs and competing uses.

Renewable is not the same as unlimited

FAO frames soil as effectively non-renewable on human planning horizons because formation is slow relative to loss. Soil functions can improve, but this is not a promise to rapidly replace lost depth. Preventing erosion and preserving productive capacity deserve priority over claiming that every damage can be repaired later. [63]

USGS groundwater analysis rejects recharge alone as a sufficient sustainable-yield rule. Pumping can alter storage and discharge as well as intercept recharge. A water budget needs the effects on connected users and ecosystems, not just an annual rainfall total. [64]

FAO’s solar-irrigation assessment warns that cheaper pumping can encourage greater extraction. Changing the energy source therefore does not establish a sustainable water withdrawal. Renewable energy and water governance solve different parts of the problem. [65]

Phosphorus management illustrates why a favorable nitrogen story can still fail. Manure has nutrient ratios that may not match crop demand; using enough to supply one nutrient can

accumulate another and increase runoff risk. A separate phosphorus ledger is necessary rather than an all-purpose fertility score. [66]

FAO’s cropland nutrient balances track inputs and removals over time and geography. National surpluses or deficits are not diagnoses for every individual farm. A system should retain both spatial scale and the specific nutrient being counted. [67]

An original accounting framework

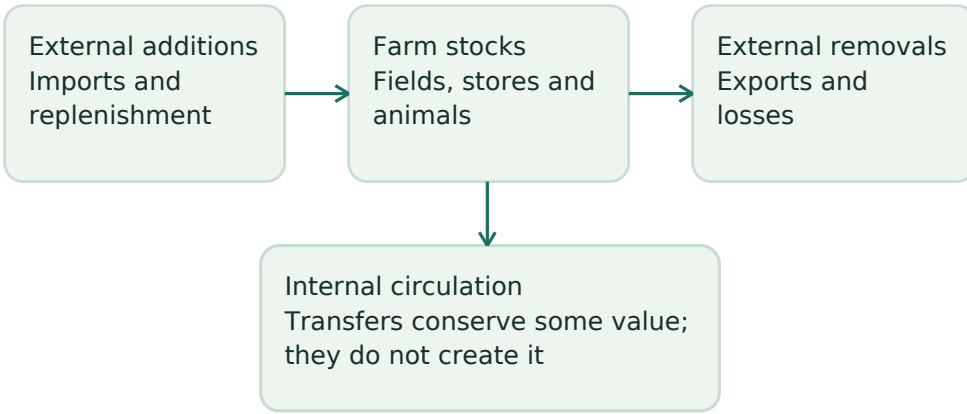
Design reading

Use a stock ledger for quantities that persist and a flow ledger for movements per interval. For a given nutrient, ending stock equals starting stock plus imports and new additions, minus exports and losses. Internal transfers between field, feed store, animal and manure store cancel at the farm boundary. Calling those transfers recycling does not turn them into new nutrients.

Track six budgets independently: nutrients by element; water by source and season; biomass by competing use; energy by service and conversion; labor by skill and critical window; and material stocks by lifetime and replacement. These categories should be linked, not collapsed into a single sustainability currency. A low-carbon machine may still consume scarce water; a productive crop may still create an impossible harvest workload.

Original toy example: a fictional field yields 100 arbitrary units of residue. Allocating 40 to animal feed, 20 to bedding and 40 to surface cover exhausts the available stock. If 15 later return as recoverable manure material, they are a delayed transformed flow, not 15 extra units of the original residue.

These numbers illustrate accounting only and are not empirical conversion coefficients.



Original accounting sketch. Internal circulation can reduce losses, but external exports, losses and replenishment still cross the system boundary.

How to recognize a successful system

Design reading

Judge the whole cycle and a difficult year, not only the showcase harvest. Ask whether the system supplies adequate usable food, protects future productive capacity, avoids shifting depletion elsewhere, supports animal welfare, and leaves workable demands on people. Include storage losses, purchased feed, imported compost, machinery replacement and access to markets or household food.

FAO Conservation Agriculture defines an integrated approach involving limited soil disturbance, soil cover and crop diversification. It is a particular framework, not a synonym for every sustainable farm. Evaluating the package requires attention to local constraints and interactions. [68]

A depth-aware synthesis of no-till studies found that climate and soil characteristics affect where carbon storage occurs. A

surface increase is not automatically an equal whole-profile gain. Carbon accounting therefore needs depth, time and uncertainty, not only a favorable topsoil sample. [69]

Design reading

Original priority proposal: prevent avoidable loss first; use appropriate diversity and biological services; recycle safe materials without double-counting; match herd and crop demand to available resources; replenish unavoidable deficits responsibly; then optimize energy and equipment. A rigid zero-external-input rule can undermine nutrient balance or welfare. The objective is durable performance with transparent costs, not symbolic purity.

CHAPTER 05

Animal husbandry as continuing care

Animals are living populations with needs, behavior and welfare outcomes, not self-maintaining converters of grass into products.

The common foundation

WOAH’s welfare framework evaluates health, comfort, nutrition, safety, pain, fear and opportunities for motivated behavior. Milk, growth or feed availability alone cannot prove good welfare. Observation should include movement, resting, injuries, body condition, responses to handling and actual access to resources. Skilled handling and suitable social conditions are part of care, not optional productivity bonuses. [70]

Animal genetic resources need to fit climate, feed, disease challenges, labor and care capacity. FAO emphasizes diversity and production context rather than one universally best breed. A useful objective includes survival, fertility, longevity, welfare and resource requirements alongside output. [71]

Biosecurity planning considers new animals, other animal contact, visitors, vehicles, tools, feed and water as possible pathways. APHIS’s small-ruminant guidance supports prevention and early recognition rather than waiting for production to collapse. Actual isolation, vaccination and treatment decisions require species-, disease- and locality-specific professional guidance. [72]

Identification and movement records support traceability when disease or food-safety problems arise. Records should connect the animal or batch with its source, movements and relevant events, rather than keeping an unexplained production total. [73]

Grazing depends on forage growth and recovery, weather, stocking and animal needs. Ground cover, infiltration and soil condition matter alongside consumption. A fixed calendar cannot guarantee recovery, particularly when drought changes growth. [74] [75]

Crop-livestock integration can use residues as feed and manure as crop nutrients, but it also moves nutrients between places and loses some along the way. Imported feed can support an apparent on-farm loop while drawing resources from elsewhere. Manure runoff can carry nitrogen and phosphorus into water. [76] [77]

Ruminant digestion emits methane, and livestock-system emissions also include manure, feed production and energy. Lower emissions per unit of product do not automatically imply lower total emissions if herd size or total output rises. Keep the denominator and system boundary explicit. [78] [79]

Species differences that a generic livestock node misses

Cattle: nutrition and water needs vary with age and production stage; calves are not small adult grazers. Comfortable resting, resource access and quiet handling matter. Breeding readiness includes physical maturity and parent-size compatibility, with pregnancy and birth risks requiring qualified support. A cattle welfare chapter is not a universal breeding calendar for other animals. [80]

Sheep and goats: feeding preferences can complement one another, but care is not interchangeable. Sheep are vulnerable to excess copper, and important parasites are shared between sheep and goats, so mixing them does not automatically break parasite cycles. Local forage and species-specific nutrition remain essential. [81]

Goats: browsing is not permission to feed arbitrary plants or refuse. Balanced nutrition, secure fencing and dry, ventilated shelter matter, with requirements changing through growth, pregnancy and lactation. The availability of browse is only one part of a viable care system. [82]

Pigs: adequate feed does not replace thermally suitable shelter, comfortable rest, exploration and foraging opportunities. Social mixing and competition can affect welfare even when aggregate food provision appears sufficient. A bare enclosure can fail behavioral needs while still producing weight gain. [83]

Laying chickens: balanced life-stage nutrition, clean water, ventilation, weather and predator protection, roosting and nesting need explicit provision. Scraps or scratch grains are not a complete diet. Guidance for small laying flocks should not be treated as a complete broiler, duck or turkey system. [84]

Horses: forage, water, shelter, safe movement, compatible companions and manure management belong together. Requirements change with work and life stage. Horse evidence is not a donkey-specific load or nutrition prescription, and body size alone does not determine safe work capacity. [85] [86]

Farmed fish: water is the life-support environment, not merely a drink. Oxygen, temperature, pH and waste-related

conditions require species- and stage-specific attention. Stocking and handling must fit the particular fish. Terrestrial livestock rules cannot be copied into an aquaculture tank. [87]

What the proposed model should preserve

Design reading

Original proposal: each animal group has care obligations, environmental tolerances, resource access, health observations, reproductive state and relevant records. Output is one outcome among several. A shortage should not silently reduce a production number while leaving welfare marked as satisfactory. Contingency capacity must include how water, feed and essential care continue during a power, labor, transport or weather disruption.

The model should also distinguish a species, a managed herd, a trained working animal and an accessible service. Knowing that donkeys exist does not supply a trained pack animal; owning one does not provide fodder along a route; transport demand does not erase welfare limits. These are interactions with the mobility chapter, not separate decorative animal traits.

CHAPTER 06

Genetic taxonomy without a ladder

Names, evolutionary relationships, managed identities and measured genetic variation answer different questions.

Four kinds of biological record

Taxonomy investigates and classifies organism groups. Nomenclature governs scientific names; the ICZN explicitly distinguishes naming rules from decisions about classification. Store a taxon identifier, accepted name, authority, synonyms and reference date separately from any genetic measurements. [88]

A phylogeny proposes evolutionary relationships and common ancestry. Its tips are not stages on a progress ladder, and rotating branches does not change their relationships. A useful display distinguishes descent from superficial visual similarity and makes uncertainty visible. [89]

A livestock breed is a recognized within-species management and identity category, informed by features such as history, geography and morphology. It need not be genetically isolated or uniform. A sampled genetic population also need not coincide exactly with a breed name or administrative boundary. [90]

A cultivar is a cultivated-plant category maintained for distinguishing characteristics under suitable propagation. It is not synonymous with a species, botanical variety or commercial brand; seed-propagated cultivars need not be

genetically identical individuals. The cited definition is from the 2016 ninth edition of the cultivated-plant nomenclatural code. [91]

Variation and boundaries

Species delimitation draws on evidence such as reproduction, morphology, ecology and evolutionary lineage. The lineage framework distinguishes the species concept from the evidence used to identify a particular boundary. No one DNA-distance threshold or ability-to-hybridize test is an infallible classifier across all organisms. [92]

Inherited variation changes through mutation, recombination, selection, migration and mating patterns. Genetic drift is random allele-frequency change and can reduce variation, especially in small isolated populations. A genetic cluster therefore needs its sampling and method recorded before being treated as a natural permanent category. [93] [94]

FAO breeding guidance connects objectives, records, production environments, relatedness and conservation. Observed performance cannot automatically be assigned to genes when feeding and management differ. Narrowing reproduction too heavily can reduce future options; evaluate adaptation, fertility and longevity alongside immediate output. [95]

The FAO global assessment treats animal genetic diversity as a resource for present use and future resilience, with threats and conservation capacity varying across contexts. Conservation and sustainable use are complementary goals rather than a choice between preserving every population unchanged and maximizing one trait. [96]

Three concrete genetic examples

A 2014 cattle study analyzed genome-wide markers in 1,543 animals from 134 bovid breeds and found major population structure together with extensive admixture. A strictly branching breed genealogy can therefore miss gene flow. The study does not establish an immutable pure-lineage hierarchy or a universally optimal breed; sampling and marker ascertainment limit the historical inference. [97]

The maize-mexicana study reported reciprocal, asymmetric gene flow and qualified evidence for highland adaptive introgression. Gene flow can coexist with recognizable morphology, but this does not establish universal benefit or safety of crossing and translocation. A linked correction was noted in the source review but its text was not retrieved, so this atlas avoids more detailed or quantitative claims from that paper. [98]

The 2018 bread-wheat reference genome represents a hexaploid genome with A, B and D subgenomes. Crop models cannot assume every organism has one simple diploid architecture. A reference genome is also not a complete representation of all wheat diversity, and additional chromosome sets do not provide a general quality or complexity ranking. [99]

Domestication is a historical process

Taming concerns an individual’s learned behavior, while domestication is considered across populations and generations. An approachable captive animal is not sufficient evidence of domestication. An evolutionary review makes that distinction without requiring every domestication history to follow an identical path. [100]

Domestication research combines archaeological context, morphology and genetics while debating the roles of control, mutualism and coevolution. A 2025 perspective instead emphasizes adaptation to human-created niches. These are active conceptual differences, not a settled universal definition or proof that intentional breeding explains every origin. [101] [102]

Comparative archaeological and genetic analysis documents recurrent wild-domestic gene flow and limits simple directed-breeding accounts of early domestication. Modern ancestry can reflect later mixing and dispersal, not only the first domestication event. Observed specimens and sequences should remain separate from inferred dates, places and intentions. [103]

Design reading

Original atlas schema: maintain separate records for taxon, managed population, individual or sample, genomic observation, trait observation and historical claim. Link them with named relations such as belongs to, sampled from, descends from or shows evidence of gene flow. Never turn a taxonomic rank, ancestry proportion or mobility tradition into a ranking of human worth or an automatic technology unlock.

CHAPTER 07

Mobility settlement and material transport

There is no single conversion from nomadic ancestry to settled civilization. Households, herds, goods and skills can move on different schedules.

Separate the moving things

Design reading

The phrase nomadic ancestry combines two distinct questions: how ancestors lived, and how a present or past population moved. Neither can be read directly from a transport tool. For this atlas, track residential mobility, seasonal herd movement, task journeys, exchange networks and population history separately. A permanent town can depend on mobile herders and merchants; a mobile household can depend on fixed stores and repeated destinations.

At Ain Mallaha, mouse communities indicate long-term forager settlement before farming, with changing occupation intensity and renewed mobility. The proxy does not reconstruct every household’s movements, but it contradicts an irreversible rule that domesticated cereals must come first and permanent settlement must follow. [104]

Dhra’s granaries, approximately 11,300-11,175 calibrated years BP, occur before morphologically domesticated crops. Their architecture is interpreted as aiding ventilation and rodent protection. Predomestication does not mean no cultivation: the study connects storage with deliberate

wild-cereal cultivation. Storage engineering and domesticated crop form are different capabilities. [105]

At Guila Naquitz, directly dated domesticated squash remains occur in a sequence of short seasonal occupations, approximately 10,000-8,000 calendar years ago. The case shows that cultivation need not await fully permanent village residence, without revealing every residence used by the community. [106]

Material culture does not identify one lifestyle

Residues in early Jomon pottery, approximately 15,000-11,800 calibrated years BP, often indicate aquatic-food processing. Ceramic cooking is not intrinsically a cereal-farming invention. Sampled vessel use is not the whole diet, and a pottery sherd alone does not prove permanent settlement. [107]

Sannai Maruyama, approximately 3900-2200 BCE, includes dwellings, storage, roads and large structures within northern Jomon hunting, gathering and fishing traditions. A substantial built settlement need not follow a cereal-state template. This does not mean there was no plant management or that all Jomon communities were identical. [108] [109]

Central Eurasian seasonal camps contain Southwest and East Asian crops by the early third millennium BCE; later camps show cultivation integrated with herding economies. Mobile pastoralists and cultivators were not mutually exclusive categories. A grain alone could be imported, so local cultivation requires wider context. [110]

A western Tianshan study reconstructs seasonal pasture circuits using archaeology, modern environmental models and

ethnographic comparison. It supports organized complementary destinations rather than random wandering, but is not direct tracking of Bronze Age people or proof that ancient access rights matched modern ones. [111]

Transport predates cities and changes production

At Kenya’s Sibilo School Road Site, with a minimum age around 200,000 years, sourced obsidian came from approximately 25, 140 and 166 km away. Material movement long predates agriculture and cities. These source distances are not one person’s journey, a known route or proof of a monetary market. [112]

Olorgesailie evidence around 320,000 years ago includes imported obsidian and on-site working debris. Raw material, prepared cores and finished tools imply different transport and production choices. Social-network explanations are interpretations; straight-line distances do not measure travel burden over rugged terrain. [113] [114]

The Malta study reports hunter-gatherer occupation beginning around 8,500 years ago, requiring roughly 100 km sea crossings under reconstructed geography. It does not reveal the exact vessel, payload or journey frequency. A January 2026 correction revises regional dating uncertainties and models without meaningfully changing the authors’ core conclusion. [115] [116]

At Abydos about 5,000 years ago, ass skeletons show load-carrying-related pathologies while retaining some proportions closer to wild asses. Transport use could precede fully modern domestic morphology. This does not specify loads, routes or a universal domestication date; trained

animals, handlers, fodder and care remain distinct requirements. [117]

The Ljubljana Marshes wheel and axle, approximately 3350-3100 BCE, combine ash and oak in a crafted assembly. The museum interprets it as a cart, potentially for timber. This is evidence of material selection and fitting parts, not proof that wheels outperform boats, carrying or pack animals on every terrain. [118] [119]

The Uluburun cargo included approximately ten tons of copper and one ton of tin. Metallurgical analysis indicates multiple processing stages, including further purification needed before casting. Bulk maritime transport could connect specialized operations, but one cargo does not establish every trader, destination or contested metal source. [120]

Networks can reorganize without losing knowledge

Portus supplied Rome through quays, warehouses, canals and ship-related infrastructure. Later trade contraction coincided with infrastructural change and more locally sourced diets. A city’s stable location depended on moving supplies; its changing provisioning cannot be explained simply by losing knowledge of ships. [121] [122]

In the twelfth-century eastern Mimbres region, smaller residential sites continued and grew while large villages lost population. Village abandonment could be regional reorganization rather than disappearance or regression to an earlier stage. This local result does not mean every transition was painless or voluntary. [123]

At Catalhoyuk, bioarchaeological evidence across roughly 7100-5950 calibrated BCE indicates changing health and

workload burdens, including increased mobility related to herding and resource acquisition. Settlement density did not mean everyone traveled less, and increased production did not guarantee improved welfare for every resident. [124]

The Qhapaq Nan network joined towns and production centers through roads, bridges, stairs, drainage, storage and accommodation. Its greatest expansion was in the fifteenth century, building on earlier routes. A transport network is a maintained landscape service, not one vehicle invention or uniform speed multiplier. [125]

An original transport decision model

Design reading

For each material, distinguish source stock, prepared material, semifinished product, finished good and repairable scrap. Record mass, bulk, fragility, perishability and the equipment needed to load or unload it. Compare labor, provisions, wear, waiting, expected loss and access obligations separately. A short route can be harder than a long one; a fast vessel is useless without a suitable landing.

At relocation, give households several options: carry, cache, leave in a maintained installation, share, rebuild or abandon. Heavy equipment can encourage return visits without requiring permanent residence. Improved transport can support a larger city or a more mobile life. These are conditional design possibilities, not laws that material weight forces one social destiny.

CHAPTER 08

A proposed Blackwish capability schema

The research can inform a richer model without being mistaken for a finished or validated one.

Node types and links

Design reading

Original proposal: use ten node types: knowledge claim, skill, tool or facility, material stock, energy source, ecological process, organism or population, institution or permission, service capability, and measurement or evidence. Keep small visible summaries while allowing each node to carry context and uncertainty. A node can be shared across multiple pathways rather than copied into isolated era branches.

Use explicit link types. Requires means necessary for a specified method, not all imaginable methods. Alternative provides another way to satisfy a requirement. Consumes and replenishes refer to quantities and time. Maintains is recurrent. Constrains captures capacity, welfare, permission or ecological limits. Historically precedes is not causal necessity. Descends from is not an unlock. Evidence for connects a claim to its support, not to a guaranteed outcome.

States and time

Design reading

A capability record should distinguish known, demonstrated, locally producible, accessible to this actor, operational, maintainable and suitable for the present goal. Add evidence status and authorization status separately. Being technically possible does not mean it is permitted, chosen or beneficial. Dormant knowledge may survive the loss of equipment; an imported service may be accessible without local manufacturing mastery.

Time belongs inside the model: season, lead time, recovery interval, service life, replacement interval, depletion, replenishment and expiry of access. A perennial forage phase, a repaired road and a breeding population cannot all update on the same instant research tick. A cycle can be reversible while its switching costs remain asymmetric.

Four text only tests

Original toy test

Dry stream. A fictional village knows a water-mill design, but the stream is dry. Knowledge remains true while operation is unavailable. Another power source is an alternative recipe with its own requirements, not a free exception.

Nutrient export. A fictional farm returns manure but exports harvest nutrients. The accounting test should reject the claim of a closed loop unless external replenishment and losses are actually balanced. Recycling is a flow, not proof of sufficiency.

Portable seeds. A fictional mobile household can carry seeds but not a large storage vessel. Plant knowledge, planting material and storage capacity stay separate. Caching,

rebuilding or exchanging storage services may be alternatives.

Closed workshop. A fictional steam city owns advanced engines but cannot legally access a repair workshop. Technical design, ownership and available service differ. The model must show an access bottleneck without declaring that engineering knowledge vanished.

Research still needed before a working simulation

Design reading

Select a particular geography, period and purpose before assigning numerical parameters. Obtain local evidence for yields, labor, transport capacity, recovery times and losses. Preserve the source version and uncertainty. Test the model against counterexamples, not only the histories it was designed to reproduce. A good first evaluation asks whether it can represent several viable systems without making them all converge on one preferred endpoint.

Keep gameplay values visibly separate from measured values. If several burdens are converted to a single score, show the chosen weights as design judgments. Evaluate aggregate output alongside distribution, care, resilience and user goals. This atlas proposes an investigable structure; it does not establish validated economics, universal historical causation, field-tested agriculture or ratified Blackwish policy.

Source register

Sources support the nearby bounded passages, not every possible claim about their subject. Some research was available through abstracts or institutional records; detailed scope and access notes remain in the companion research records. Game counts refer to the cited version or ruleset, not an invariant total.

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99. IWGSC: Shifting the limits in wheat research and breeding using a fully annotated reference genome
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100. Driscoll, Macdonald and O'Brien: From wild animals to domestic pets, an evolutionary view of domestication
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101. Zeder: Core questions in domestication research
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102. Lord et al.: A universally applicable definition for domestication
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103. Marshall et al.: Evaluating the roles of directed breeding and gene flow in animal domestication
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104. Origins of house mice in ecological niches created by settled hunter-gatherers in the Levant 15,000 y ago
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106. The Initial Domestication of Cucurbita pepo in the Americas 10,000 Years Ago
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107. Earliest evidence for the use of pottery
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108. Sannai Maruyama Site
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109. Jomon Prehistoric Sites in Northern Japan pamphlet
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114. Olorgesailie Drilling Project
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115. Hunter-gatherer sea voyages extended to remotest Mediterranean islands
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118. Replica of the oldest wooden wheel with an axle for the winner of the 26th Tour of Slovenia
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119. City Museum of Ljubljana
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121. Portus: The Maritime Dimension of Imperial Rome
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124. Bioarchaeology of Neolithic Çatalhöyük reveals fundamental transitions in health, mobility, and lifestyle in early farmers
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125. Qhapaq Ñan, Andean Road System
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